



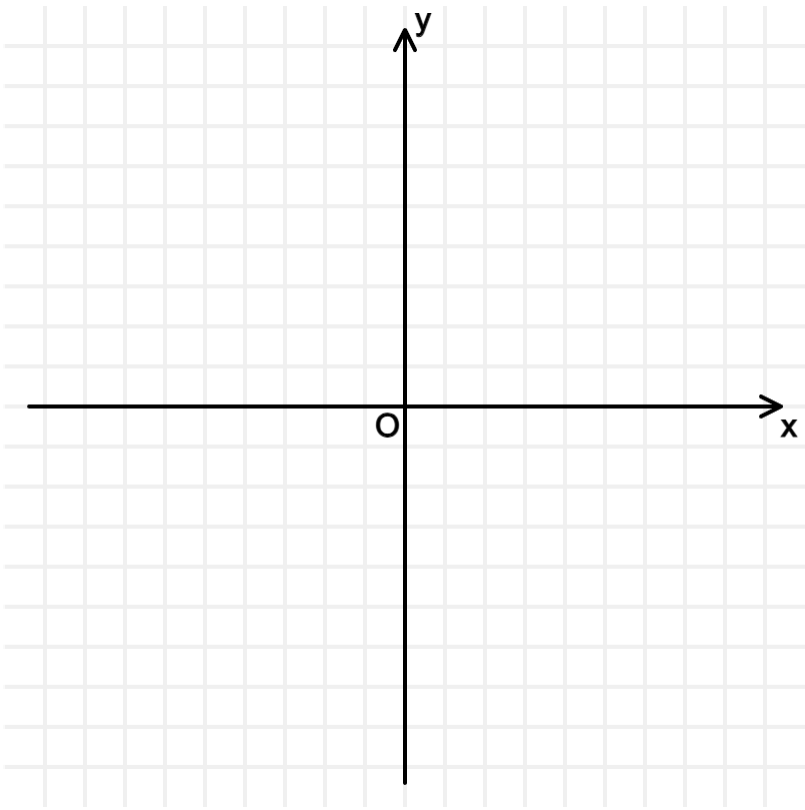
Exponentials and Logs

Subject & topics: Maths Stage & difficulty: A Level C2

Part A

Sketching

Sketch the curve $y = 6 \times 5^x$.



Part B

Find y -intercept

What is the value of the y -intercept of the curve $y = 6 \times 5^x$?



The following symbols may be useful: y

Part C

Find x -coordinate

The point P on the curve $y = 9^x$ has y -coordinate equal to 150. Use logarithms to find the x -coordinate of P.

Give the x -coordinate of P to 3 significant figures.

Part D

New x -coordinate

The curves $y = 6 \times 5^x$ and $y = 9^x$ intersect at the point Q.

Find the exact value of the x -coordinate at point Q, giving any logarithms in base three ($\log_3$).

When you are entering your answer, note that $\log_a b$ can be written using `log(b,a)`.

The following symbols may be useful: `log()`, `x`

Used with permission from UCLES, A Level, January 2010, Paper 4722, Question 9.

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Solving Equations & Logs 2i

Subject & topics: Maths Stage & difficulty: A Level P2

Part A

Solve equation

Use logarithms to solve the equation $2^{n-3} = 18000$, giving your answer to 3 significant figures.

Part B

Simultaneous equations

Solve the simultaneous equations $\log_2 x + \log_2 y = 8$ and $\log_2(\frac{x^2}{y}) = 7$

(,)

Used with permission from UCLES, A Level Maths, June 2015, OCR C2, Question 8

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Logarithmic Plots 4

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level P2

A student used a graph of $\ln y$ against x to discover that $y = e^{2x+5}$.

What were the gradient and intercept of the graph?

Part A

Find the gradient

What was the gradient of the graph?

Part B

Find the intercept

What was the intercept of the graph?

Adapted for Isaac Physics from NST IA Biology preparation work

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Straight Lines: Gradients and Normals 1i

Subject & topics: Maths **Stage & difficulty:** A Level P1

Part A

Gradient of line

Find the gradient of the line l_1 which has equation $4x - 3y + 5 = 0$.

Part B

Perpendicular line

Find the equation of the line l_2 , which passes through the point $(1, 2)$ and is perpendicular to the line l_1 , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers.

The following symbols may be useful: x , y

Part C

Midpoint

The line l_1 crosses the x -axis at P and the line l_2 crosses the y -axis at Q . Find the coordinates of the midpoint of PQ . State your answer using exact decimals.

(,)

Part D

Length of PQ

Find the length of PQ .

Used with permission from UCLES, A level, June 2005, Paper 4721, Question 9

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Algebra and Roots: Cubics with Substitution 1ii

Subject & topics: Maths **Stage & difficulty:** Further A P1

The cubic equation $3x^3 - 9x^2 + 6x + 2 = 0$ has roots α , β and γ .

Part A

Roots

Give the value of $\alpha + \beta + \gamma$.

$$\alpha + \beta + \gamma = \text{[input box]}$$

Give the value of $\alpha\beta + \beta\gamma + \gamma\alpha$.

$$\alpha\beta + \beta\gamma + \gamma\alpha = \text{[input box]}$$

Part B

$$\alpha^2 + \beta^2 + \gamma^2$$

Hence, find the value of $\alpha^2 + \beta^2 + \gamma^2$.

Part C

Substitution

Use the substitution $x = \frac{1}{u}$ to find a cubic equation in u in the form $au^3 + bu^2 + cu + d = 0$ where a, b, c and d are integers.

The following symbols may be useful: u

Part D

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

Adapted with permission from UCLES, A Level, June 2007, Paper 4725, Question 6.

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Roots of Polynomials

Subject & topics: Maths **Stage & difficulty:** A Level C2

This question is about manipulation of the roots of two polynomials.

$$x^2 + kx + 2k = 0$$

has the roots α and β , while

$$x^3 + 4x + 3 = 0$$

has the roots α' , β' and γ' . Take $k \neq 0$.

Part A

Roots of the quadratic

Find a quadratic equation with roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

The following symbols may be useful: k , x

Part B

Substitution

Starting from the cubic equation above, use the substitution $x = \sqrt{u}$ to obtain a cubic equation in u .

The following symbols may be useful: u

Part C

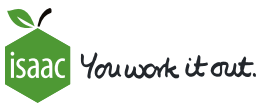
Roots of the cubic

Find an expression for $\alpha'^4 + \beta'^4 + \gamma'^4 + \alpha'\beta'\gamma'$.

Adapted with permission from UCLES, A Level, June 2008, Paper 4725, Question 8 and June 2015, Paper 4725, Question 10.

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Matrices: nxm Rules 1i

Subject & topics: Maths Stage & difficulty: Further A P2

The matrices **A**, **B** and **C** are given by $\mathbf{A} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 2 & 4 & -1 \end{pmatrix}$

Part A

A – 4B

Find **A – 4B**.

(

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Part B

BC

Find **BC**.

(

)

Part C
CA

Find the matrix **CA**.

(

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Adapted with permission from UCLES, A Level, June 2008, Paper 4725, Question 5.

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Matrices

Subject & topics: Maths **Stage & difficulty:** Further A C2

This question will look at 4 matrices **A**, **B**, **C** and **D** given by

$$\mathbf{A} = \begin{pmatrix} 3 & 4 \\ 2 & -3 \end{pmatrix}$$

$$\mathbf{B} = \begin{pmatrix} 4 & 6 \\ 3 & -5 \end{pmatrix}$$

$$\mathbf{C} = \begin{pmatrix} a & 3 \\ -2 & 1 \end{pmatrix}$$

$$\mathbf{D} = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 3 & 1 \\ 1 & 1 & b \end{pmatrix}$$

where $b \neq 1$.

Part A

Matrix equation

p and q satisfy the equation $p\mathbf{A} + q\mathbf{B} = \mathbf{I}$ where $\mathbf{I}$ is the identity matrix.

Find the constants p and q . If a value is not a whole number, enter the value as a decimal.

$p =$

$q =$

Part B

A singularity!

Given that $\mathbf{C}$ is singular, find the value of a .



The following symbols may be useful: a

Part C

$\mathbf{C}^{-1}$

Given that $\mathbf{C}$ is non-singular, find $\mathbf{C}^{-1}$.

$$\mathbf{C}^{-1} = \frac{1}{} \begin{pmatrix} & \\ & \end{pmatrix}$$

Items:

- 3

-2

-1

0

1

2

3

-a

a

a + 2

a + 3

a + 6

a - 2

a - 3

a - 6

Part D

Simultaneous equations

Using the previous part, solve the following simultaneous equations

$$ax + 3y = 1$$

$$-2x + y = -1$$

(,)

Items:

- 6

-4

-2

0

2

4

6

-a

a

a + 2

a + 4

a + 6

2 - a

4 - a

6 - a

Part E

|D|

Find the determinant of **D** in terms of *b*.

The following symbols may be useful: b

$$\mathbf{D}^{-1}$$
Find $\mathbf{D}^{-1}$.

$$\mathbf{D}^{-1} = \frac{1}{\boxed{}} \begin{pmatrix} \boxed{} & \boxed{} & \boxed{} \\ \boxed{} & \boxed{} & \boxed{} \\ \boxed{} & \boxed{} & \boxed{} \end{pmatrix}$$

Items:

[illegible]

More simultaneous equations

Using the answer to the previous part, or otherwise, solve the following set of simultaneous equations

$$2x - y + z = 1$$

$$3y + z = 2$$

$$x + y + bz = 2$$

$$\left(\begin{pmatrix} \\ \end{pmatrix}, \begin{pmatrix} \\ \end{pmatrix}, \begin{pmatrix} \\ \end{pmatrix} \right)$$

Items:

-2 -1 0 1 2 $-b$ b $2b-2$ $2b-5$ $2b-6$ $4b-5$ $4b-6$ $4b-7$ $5b-5$ $5b-6$
 $5b-7$ $6b-5$ $6b-6$ $6b-7$

Adapted with permission from UCLES, A Level, June 2012, Paper 4725, Question 2, UCLES, A Level, June 2008, Paper 4725, Question 7 and UCLES, A Level, June 2010, Paper 4725, Question 9

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